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Geomagnetic paleosecular variation at Hawaii around 3 Ma from a sequence of 107 lava flows at Kaena Point (Oahu)

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Abstract

We have sampled a sequence of 107 lava flows in the Waianae series on Oahu, Hawaii, in two separate stretches. The first (51 flows) at Kaena Point extends from sea level to an altitude of 190 m, and the second (56 flows) follows the Satellite Tracking Station road from 218 m to 360 m above sea level. Thermal and af demagnetization yield very similar results, and reveal only normal polarities. K/Ar dating at five different horizons combined with the paleomagnetic results indicate that the sequence was emplaced entirely in the normal polarity interval between the Upper Mammoth and the Lower Kaena polarity transitions (3.22–3.11 Ma). Some of the flows are serially correlated. Filtering does not, however, drastically affect the final results. Inclinations are shallower than those expected from a centered dipole field. Although large (∼13º), the inclination anomaly is consistent with results from other sites at the same latitude. The scatter of the VGPs about the geographic pole is consistent with available data from other regions at similar latitudes. The scatter of the directions is also consistent with the predictions of recent statistical models of paleosecular variation. Therefore, these results do not support the hypothesis of a Pacific dipole window. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: magnetic field; secular variations; K/Ar; absolute age; Hawaii

1. Introduction

The hypothesis that the central Pacific might have been a region of low non-dipole geomagnetic field was suggested by Cox and Doell on the basis of the small angular dispersion observed in their extensive data set from the Hawaiian Islands [1–3]. This suggestion agreed with an earlier observation [4] that the historical rates of change of the geomagnetic field were particularly low in the central Pacific region.

Subsequently, Doell himself [5–7] noted that some of the results from successive flows exhibited serial correlation, so that they could not be considered as independent observations. In their study of radiocarbon dated flows on the island of Hawaii Coe et al. [8] found paleointensities for several of them much higher than the worldwide average for those times and normal angular dispersion, which they interpreted as signaling the presence of sizable non-dipole fields. They reanalyzed the data from all the Hawaiian Islands and concluded that the amplitude of secular variation only appeared low because much of the data had been obtained from sequences that erupted so rapidly that the secular variation had

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been incompletely sampled. Tanaka and Kono [9] and Mankinen and Champion [10,11] have also obtained data supporting the existence of a significant non-dipole field in Hawaii.

Recently, McElhinny et al. [12] have re-examined the old data from Hawaii and Tahiti and used a new statistical method to take proper account of those flows which have repeatedly sampled the same geomagnetic field vector. They concluded that the amplitude of the secular variation at Hawaii is consistent with values from other worldwide scattered sites, so that the hypothesis of the Pacific dipole window may confidently be rejected.

We report here on new paleomagnetic results obtained from a sequence of 107 flows of the Waianae volcano in the northwestern part of Oahu which was emplaced around 3 Ma. Results from the Waianae range have already been reported [13]. However, although extensive, these previous results were obtained from relatively short sequences, and the possibility of time overlaps or gaps between them cannot be excluded. In this study, we have studied a single long sequence characterized by unambiguous stratigraphy, so that sampling artifacts due to imperfect correlation and possible overlaps are avoided. In addition, it may be worthwhile to repeat this study from an important locality such as Hawaii, in the middle of the Pacific Ocean, because some of the methods used in the previous studies done over 30 years ago have since been superceded by modern laboratory methods [14].

2. Geological setting and sampling

The Waianae volcano is the more western and older of the two volcanoes that form the island of Oahu in Hawaii. Early K/Ar dating [13] indicated that the subaerial volcanic activity of the Waianae volcano occurred in the interval 3.8 to 2.4 Ma. More recent age determinations [14,15], some of which are discussed below, have confirmed these broad trends of the Waianae volcanic history.

After a first shield-building stage (Lualualei member) dominated by the eruption of tholeiitic basalts [16], the second shield-building stage (Kamaileunu member) is characterized by an increasing variability in lava composition, comprising plagioclase-phyric tholeiitic basalts, alkali olivine basalts and plagioclase-phyric basaltic hawaiites. Eruptions of the Kamaileunu lavas occurred within the caldera and along the rift zones outside the caldera during the Gauss Normal Polarity Chron [16]. The post-shield stage started with the Palehua member which consists of hawaiites and to a lesser extent of mugearites [17,18].

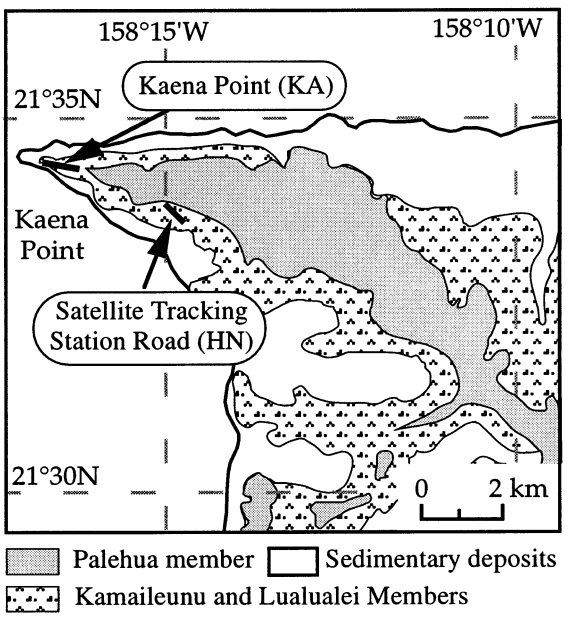


Fig. 1. Schematic geological map of the Waianae volcano (adapted from [17]) showing the position of the KP–STSR sequence.

The sequence investigated in this study is located at the northwestern termination of the Waianae ridge (lat. 21.4ºN; long. 158.2ºW) and consists of two sections: a lower one, sampled at Kaena Point (KP), is comprised of 51 successive flows extending from sea level to an altitude of 190 m; the second, comprised of 56 flows, follows the Satellite Tracking Station Road (STSR) just south of KP, between 218 and 353 m (Fig. 1). The two sections form a single sequence of 107 flows. The two top flows (HN01 and HN02) are olivine-rich basalts, with olivine crystals 1–5 mm in size, clearly showing that they belong to the Kamaileunu member. The Kamaileunu–Palehua contact is thus not encountered in the section and is higher in the stratigraphy, consistent with previous surveys [16].

On average 6–7 cores per flow, with a minimum

of 5, were sampled for this study. For each flow, sampling extended laterally at least 10–30 m, sometimes more, especially in the KP exposure. The precise location of each core was chosen after checking that no strong magnetic anomaly was present, by progressively approaching a compass to the surface of the lava, in order to avoid sampling cores at particular zones struck by lightning. The lower part of the KP section, which is a well exposed cliff, demanded greater care than the upper part of the section along the STSR. Overall, 11 cores from 8 flows (2 in the STSR exposure and the others in the KP exposure) are characterized by magnetizations higher than 25 A/m, as measured later in the laboratory and were not considered in the final results. A total of 615 cores were sampled using a gasoline-powered drill equipped with a 25-mm diamond bit. In general, the flows are well exposed and the stratigraphy is quite clear. Samples from the KP exposure were labeled KA with numbers increasing upward from the bottom (except for KA51 at the very bottom). Samples from the STSR exposure were labeled HN with number increasing from the top downward. This contrasting way of labeling reflects the order in which the flows were sampled in the field. There is no HN40 flow. Because only two cores could be obtained from the very thin HN10 flow, this flow was not considered further. The flows are exposed almost continuously along the upper part of the STSR from HN01 to HN42. Below this level the sequence is occasionally covered by vegetation and there are gaps in the sampling. The most important gap may represent 3–4 flows, based on the average thickness of 2 m per flow. At the top of the sequence, the paleomagnetic sampling ends at an altitude of 353 m because the flows above this altitude were cemented to prevent rock falls onto the road and could not be sampled.

Samples were oriented in situ using a magnetic and, in the large majority of the cases, a sun compass. Declination anomalies range within a few degrees about the mean regional anomaly (12ºE), for all but five cores in the STSR exposure. Each core typically yielded 5–6 standard 22 × 25 mm specimens, allowing laboratory studies of the magnetic mineralogy, and alternating field and thermal demagnetization of the natural remanent magnetization (NRM).

Fresh massive samples, or three unoriented cores, were sampled at six different stratigraphic horizons for K/Ar dating. The very top sample was taken at an altitude of 400 m, about 50 m above and some 500 m away from the flows with welded contacts that comprise the STSR sequence. This sample clearly belongs to the Palehua member and therefore does not belong to the sequence studied here, but provides an upper limit for the age of the underlying flows.

3. Laboratory experiments and results.

3.1. K/Ar dating

The K–Ar technique used in this study has been described elsewhere [19–21]. It has already been successfully used for dating much younger lavas from Hawaii and other hotspot seamounts [23,24]. K and Ar abundances were measured on the microcrystalline groundmass because it is not likely that this fraction, which crystallized after the eruption, contains excess argon due to incomplete degassing. Phenocrysts and xenocrysts, which may carry inherited argon, were separated from the groundmass using heavy liquids and magnetic separation, and their absence in the groundmass aliquots selected for measurements was carefully checked with a binocular microscope. Ages were calculated using the decay constants of Steiger and Jäger [22]. The reproducibility of the laboratory results can be directly checked by comparing the data in Table 1 of the two independent replicate measurements made with different amounts of aliquots. Standard errors (95% confidence level) were calculated from 100 measurements for each argon isotope.

The ages (Table 1) are all consistent with stratigraphic order and span an interval of 0.3 m.y. All attempts to date the flows of the upper part of the sequence failed because they are characterized by extremely low potassic contents. However, a reliable date of 2.97 ± 0.032 Ma was obtained for the sample taken at 400 m which belongs to the Palehua member of the Waianae volcano that is characterized by a higher K-content.

3.2. Rock-magnetic analysis

High field thermomagnetic analysis of over 200 samples (1 to 3 per flow) was done with a horizon-

Table 1
Radiometric (K/Ar) dates obtained from the lava flows of the Waianae volcano

Flow	Sample (K/Ar)	Sequence	Height a.s.l. (m)	K (wt.%)	$^{40}Ar^{*}$ (%)	$^{40}Ar^{*}$ (10^{-13} moles/g)	Age ($\pm 2\sigma$) (Ma)	Age (Ma, mean value)
–	1B	Palehua	400	1.155 ± 0.012	27.527	59.328	2.96 ± 0.03	
–	1B		400		25.595	59.736	2.98 ± 0.03	2.97 ± 0.02
HN55	HN55	STSR	220	0.415 ± 0.004	8.766	22.337	3.10 ± 0.04	
HN55	HN55		220		11.720	22.438	3.11 ± 0.04	3.11 ± 0.03
KA48	Ka7	KP	190	0.375 ± 0.004	10.564	20.746	3.19 ± 0.04	
KA48	Ka7		190		9.589	20.432	3.14 ± 0.04	3.16 ± 0.03
KA02	Ka01bis	KP	10	0.389 ± 0.004	5.796	21.812	3.23 ± 0.03	
KA02	Ka01bis		10		4.431	21.273	3.15 ± 0.03	3.19 ± 0.03
–	OAU 06	KP	0	0.312 ± 0.003	9.810	17.708	3.27 ± 0.05	
–	OAU 06		0		6.926	17.757	3.28 ± 0.04	3.27 ± 0.03

Age calculations are based on the decay and abundance constants of Steiger and Jäger [22]

tal Curie balance. Heating and cooling rates were close to 7–8ºC/min and the maximum temperature reached was of the order of 700ºC. Argon was continuously flushed into the measuring cell to limit oxidation of the magnetic minerals during the experiments.

Representative results are shown in Fig. 2. Over 3/4 of the samples are characterized by an almost reversible behaviour with Curie temperature in the 550–580ºC range, typical of magnetite or low Ti-content titanomagnetite. In some cases a second phase was observed with Curie temperature in the range 250–350ºC, which might correspond to either a higher Ti-content magnetite or maghemite. In these cases the curve is not reversible. These two main types of thermomagnetic behaviour may characterize samples from the same flow, illustrating some heterogeneity. As discussed below, even in cases where the thermomagnetic curves are complex, the characteristic remanent magnetization (ChRM) could be unambiguously isolated.

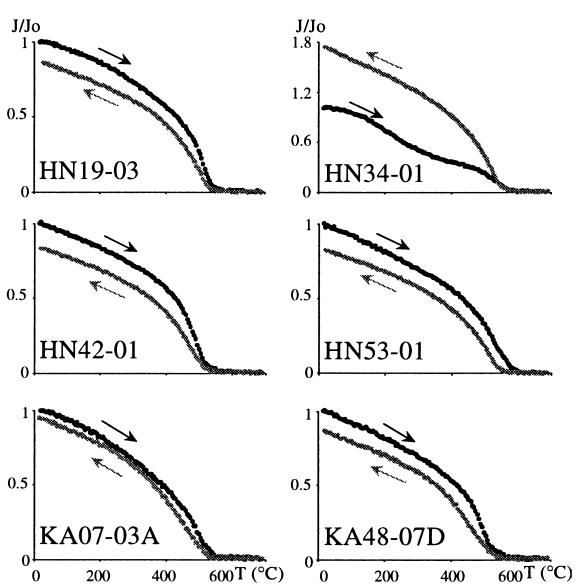


Fig. 2. Representative results for the thermomagnetic analyses for the KP–STSR sequence.

3.3. Demagnetization of the NRM

The remanent magnetization was measured with a 3-axes cryogenic magnetometer or with a spinner magnetometer in a μ-metal shielded room. NRMs range from 0.5 to 56 A/m, with a distribution peaked around 6.7 A/m. Stepwise af and thermal demagnetization isolate virtually identical paleomagnetic directions on samples from the same core. Only very occasionally, differences of up to 10–15º were observed. Thermal demagnetization usually yielded slightly 'noisier' demagnetization diagrams but allowed complete removal of the NRM, while some 5% of it was left after af demagnetization with the peak af-field of 0.1 T available in the laboratory. On the average, at least two samples per flow were af-demagnetized and three treated thermally for the STRS exposure. All the samples from the KP expo-

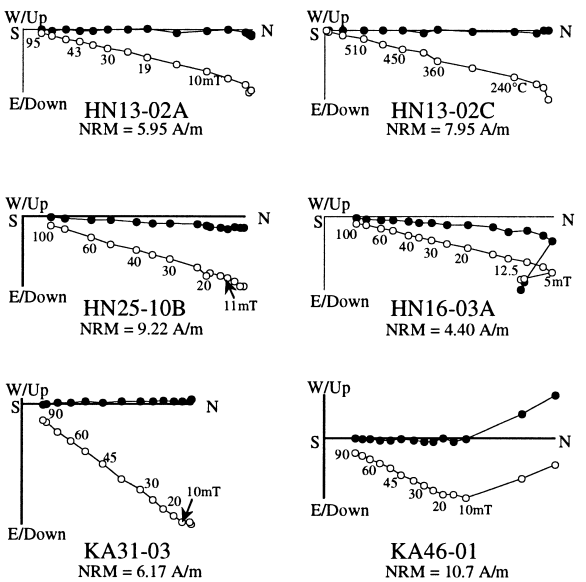


Fig. 3. Representative demagnetization diagrams for samples from the KP and STSR sections. Apart from an occasional secondary component, easily demagnetized in the very first steps of the demagnetization, a single stable component decaying to the origin was observed for a great majority of the samples.

sure were af demagnetized, because this technique is better adapted for revealing possible influence by lightning strikes. In general, the low-field magnetic susceptibility measured at room temperature after each step of thermal demagnetization documents magneto-mineralogical changes at temperatures exceeding 450ºC. These changes did not perturb the thermal cleaning because the very low residual field in the furnace (~2 nT) prevents acquisition of spurious magnetizations during the heating.

Examples of demagnetization diagrams are shown in Fig. 3. Over 80% of the samples are characterized by a single stable component of magnetization. A secondary component occasionally present in the NRM is easily removed after heating to temperatures usually lower than 150–200ºC (in some isolated cases up to 250–300ºC) or af treatment in fields lower than 15–20 mT. Principal component analysis of the diagrams yielded MAD angles of the order of 2º for the direction of the characteristic remanent magnetization (ChRM) of each sample. Samples from the same flow yielded very consistent results and the within-flow scatter is characterized by an average value of 5.0 ± 2.3º over the entire section. Exceptions are flows HN14 and KA24, for which within-flow scatter exceeds 12º. Only two samples from flow HN15 yielded acceptable demagnetization diagrams, so this flow was not considered further.

3.4. The record of geomagnetic paleosecular variation

The reliable ChRM directions are reported in Table 2 and in Fig. 4 in stratigraphic order. The results of the radiometric dating and the altitudes are also shown in the figure. Visually, declinations appear to be centered around the north–south direction while inclinations are generally shallower than that expected from a geocentered axial dipole. This feature, already reported at many geographically widespread sites (see, e.g. [25]), results in VGPs lying on the far side of the geographic pole with respect to the sampling site (Fig. 5).

Normal polarities were observed throughout the section. Two negative inclinations are recorded by lava flows KA05 and KA06bis, but the associated declinations are only slightly deviated, so that the corresponding VGPs are still in the Northern Hemisphere (the lowest, for KA05, at about 40ºN). The directions of the flows underlying KA05 are also slightly deviated with respect to their neighboring flows, and the NRM of the flows overlying and underlying flows KA05 and KA06bis are the smallest of the entire section, suggesting that this anomalous direction might represent a partial record of a geomagnetic excursion, as discussed below. A declination swing of about 35º, with the inclinations showing rather high values, is observed in the upper part of the section, but no particular trend was observed in the NRM.

4. Discussion

4.1. Chronological constraints

Table 1 shows that the K/Ar ages obtained at different horizons of the sequence are consistent with their stratigraphic order, which is a first-order check of their reliability. However, according to the dating, the Kaena subchron of reversed polarity (3.11–3.04

Table 2
Paleomagnetic results from the KP–STSR sequence

Flow no.	n/N	Dec.	Inc.	k	α_{95}	VGP (Lat.)	VGP (Long.)	n/N	Dec.′	Inc.′	k'	α'_{95}	VGP′ (Lat.)	VGP′ (Long.)	Cluster no.
STSR sequence															
HN 01	*5/5*	*5.6*	*28.7*	*386*	*3.5*	*81.8*	*−19.8*	*5/5*	*5.6*	*28.7*	*386*	*3.5*	*81.8*	*−19.9*	*1*
HN 02	5/6	6.4	16.3	555	2.9	75.4	−4.3								
HN 03	4/4	9.3	14.2	1332	2.2	73.1	−11.8	9/10	7.8	15.2	671	3.6	74.3	−8.3	2
HN 04	*4/5*	*5.6*	*10.2*	*474*	*3.7*	*72.7*	*2.5*	*4/5*	*5.6*	*10.2*	*474*	*3.7*	*72.8*	*2.5*	*3*
HN 05	5/6	34.6	47.1	613	2.8	58.0	−87.6								
HN 06	5/5	39.1	45.6	190	5.0	54.1	−85.1	10/11	36.8	46.4	336	2.6	56.1	−86.3	4
HN 07	*6/6*	*32.8*	*41*	*158*	*4.9*	*59.7*	*−78.3*	*6/6*	*32.8*	*41.0*	*158*	*4.9*	*59.7*	*−78.4*	*5*
HN 08	3/4	3.3	32.2	439	6.9	84.9	−16.6	3/4	3.3	32.2	439	6.9	84.9	−16.6	6
HN 09	*4/5*	*2.3*	*43.7*	*290*	*4.7*	*85.4*	*−131.2*	*4/5*	*2.3*	*43.7*	*290*	*4.7*	*85.4*	*−131.2*	*7*
HN 11	5/5	2.1	19.9	334	3.7	78.6	11.2								
HN 12	4/5	2	14.9	172	6.1	75.9	13.5								
HN 13	4/5	0.6	13.4	77	9.1	75.3	19.3								
HN 14	4/5	13.2	13.5	35	13.6	70.5	−21.2								
HN 16	6/6	2.9	15	263	3.8	75.8	9.8								
HN 17	4/4	1.3	19.3	210	5.5	78.4	15.3								
HN 18	5/5	1.6	16.7	578	2.8	76.9	14.6								
HN 19	6/6	1.7	18.6	109	5.9	77.9	13.6								
HN 20	5/5	1.2	19.4	226	4.6	78.4	15.8								
HN 21	5/5	2	17.7	290	4.0	77.4	12.6								
HN 22	5/5	2.9	20	158	5.5	78.5	7.3								
HN 23	5/5	3.8	15.6	274	4.1	76.0	5.9								
HN 24	5/5	2.1	19.4	1002	2.2	78.3	11.4	63/66	2.8	17.3	1676	1.4	77.1	9.3	8
HN 25	*9/10*	*1.2*	*16*	*984*	*1.5*	*76.6*	*16.5*								
HN 26	*5/5*	*2.1*	*14.8*	*602*	*2.8*	*75.9*	*13.1*	*14/15*	*1.5*	*15.6*	*960*	*1.3*	*76.4*	*15.5*	*9*
HN 27	5/5	5.4	20.8	257	4.3	78.1	−4.8								
HN 28	5/5	7.6	19	170	5.3	76.2	−11.3	10/10	6.4	19.8	254	3.0	77.1	−7.8	10
HN 29	*5/5*	*346.7*	*18.8*	*2264*	*1.4*	*72.6*	*70.9*								
HN 30	*3/5*	*343.6*	*14.9*	*217*	*6.8*	*68.9*	*72.8*	*8/10*	*−14.5*	*17.3*	*489*	*2.5*	*71.2*	*71.9*	*11*
HN 31	5/5	9.9	10.5	354	3.6	71.2	−10.3	5/5	9.9	10.5	354	3.6	71.2	−10.3	12
HN 32	*5/6*	*3.6*	*11.7*	*298*	*4.0*	*74.0*	*8.5*								
HN 33	*4/4*	*5.1*	*11.5*	*171*	*6.1*	*73.5*	*3.5*								
HN 34	*4/4*	*9.2*	*13.8*	*62*	*10.1*	*72.9*	*−11.2*	*13/14*	*5.7*	*12.3*	*166*	*3.2*	*73.7*	*0.9*	*13*
HN 35	5/6	7.1	17.3	485	3.1	75.6	−7.7	5/6	7.1	17.3	485	3.1	75.6	−7.8	14
HN 36	*4/5*	*5.6*	*26.4*	*259*	*4.9*	*80.7*	*−14.4*	*4/5*	*5.6*	*26.4*	*259*	*4.9*	*80.7*	*−14.4*	*15*
HN 37	4/5	356.1	26.5	275	4.8	81.6	48.7								
HN 38	3/6	359.5	23.6	147	8.3	80.8	24.8	7/11	−2.5	25.1	277	3.6	81.4	37.9	16
HN 39	*5/5*	*353.5*	*31.5*	*289*	*4.0*	*82.4*	*76.8*								
HN 41	*5/5*	*356.9*	*33.3*	*377*	*3.5*	*85.6*	*63.7*	*10/10*	*−4.9*	*32.4*	*374*	*2.5*	*84.0*	*72.5*	*17*
HN 42	5/5	3.9	32.0	360	3.6	84.5	−20.5	5/5	3.9	32.0	360	3.6	84.5	−20.5	18
HN 43	*5/5*	*20.7*	*29.3*	*148*	*5.6*	*69.5*	*−55.3*	*5/5*	*20.7*	*29.3*	*148*	*5.6*	*69.5*	*−55.3*	*19*
HN 44	5/5	354.9	35.6	297	4.0	85.0	92.0	5/5	354.9	35.6	297	4.0	85.0	92.0	20
HN 45	*6/6*	*15.9*	*35.3*	*215*	*4.8*	*75.0*	*−63.5*	*6/6*	*15.9*	*35.3*	*215*	*4.8*	*75.0*	*−63.5*	*21*
HN 46	5/5	354.1	29.0	279	4.1	81.8	65.7								
HN 47	4/5	355.8	29.2	736	2.9	82.9	56.5	9/10	−5.2	29.1	541	2.2	82.3	62.4	22
HN 48	*6/6*	*359.4*	*25.8*	*681*	*2.3*	*82.1*	*26.0*								
HN 49	*7/7*	*357.4*	*27.7*	*784*	*2.0*	*82.7*	*42.2*								
HN 50	*5/5*	*358.5*	*29.2*	*240*	*4.4*	*83.9*	*35.6*								
HN 51	*5/5*	*358.5*	*26.2*	*198*	*4.9*	*82.2*	*32.5*								
HN 52	*4/5*	*357.1*	*26.7*	*63.3*	*9.9*	*82.1*	*42.9*	*27/28*	*−2.2*	*27.3*	*241.5*	*1.8*	*82.7*	*39.2*	*23*
HN 53	5/5	2.7	32.4	137	5.9	85.3	−11.9	5/5	2.7	32.4	137	5.9	85.3	−11.9	24
HN 54	*5/5*	*1.8*	*43.0*	*179*	*5.1*	*86.1*	*−133.3*	*5/5*	*1.8*	*43*	*179*	*5.1*	*86.1*	*−133.3*	*25*
HN 55	4/5	3.2	25.4	316	4.5	81.3	0.6								
HN 56	6/6	359.7	24.2	222	4.1	81.2	23.6								
HN 57	5/5	0.1	26.1	114	6.4	82.3	20.9	15/16	0.75	25.0	236	2.5	81.6	16.7	26

Table 2 (continued)

Flow no.	n/N	Dec.	Inc.	k	α_{95}	VGP (Lat.)	VGP (Long.)	n/N	Dec.′	Inc.′	k'	α'_{95}	VGP′ (Lat.)	VGP′ (Long.)	Cluster no.
KP sequence															
KA 48	*8/8*	*11*	*40*	*91*	*5.5*	*79.7*	*−77.7*								
KA 47	*6/6*	*11.4*	*36*	*81*	*6.8*	*79.3*	*−62.7*	*14/15*	*11.2*	*38.3*	*102*	*3.9*	*79.6*	*−71.3*	*27*
KA 46	5/8	−2.7	27.4	207	4.8	82.7	42.4								
KA 45	7/7	−1.4	26.9	304	3.2	82.7	32.4								
KA 44	6/9	−0.9	24.2	672	2.4	81.2	27.9	18/24	−1.6	26.1	379	1.8	82.3	33.3	28
KA 43	*7/8*	*5.1*	*21.7*	*109*	*5.4*	*78.8*	*−4.7*								
KA 42	*7/7*	*1.8*	*22.0*	*293*	*3.3*	*79.9*	*11.6*								
KA 41	*6/6*	*0.4*	*25.9*	*101*	*6.1*	*82.3*	*19.2*								
KA 40	*5/6*	*4.6*	*22.3*	*168*	*5.3*	*79.3*	*−3.3*	*25/27*	*2.9*	*22.9*	*161*	*2.3*	*80.2*	*4.5*	*29*
KA 39	5/5	1	27.7	680	2.6	83.2	13.9								
KA 38	5/6	0.7	32.9	92	7.2	86.4	10.8								
KA 37	5/5	−0.1	28.2	292	4	83.6	22.6	15/16	0.5	29.6	221	2.6	84.5	16.5	30
KA 36	*5/5*	*2.8*	*21.2*	*363*	*4*	*79.2*	*7.1*								
KA 35	*6/7*	*3.3*	*23.6*	*23*	*4.6*	*80.4*	*2.3*	*11/12*	*2.8*	*23.1*	*355*	*2.6*	*80.3*	*4.9*	*31*
KA 34	8/9	2	28.9	307	3	83.7	3.6								
KA 33	6/6	−1	30.9	103	6	85.1	33.0	14/15	0.7	29.7	197	2.8	84.5	14.1	32
KA 32	*6/7*	*0.7*	*46.6*	*93*	*6.4*	*83.5*	*−152.0*	*6/7*	*0.7*	*46.6*	*93*	*6.4*	*83.5*	*−152.0*	*33*
KA 31	5/6	3.1	34.2	194	4.9	86.0	−26.6								
KA 30	6/6	−1.5	39.8	50	8.7	88.1	153.0	11/12	0.7	37.2	84	5.0	89.2	−28.8	34
KA 29	*6/8*	*−10.6*	*37.0*	*140*	*5.2*	*80.7*	*112.3*								
KA 28	*9/11*	*−2.1*	*36.5*	*60*	*6.3*	*87.8*	*83.6*	*15/19*	*−5.3*	*37.1*	*82*	*4.2*	*85.0*	*105.3*	*35*
KA 27	3/7	6.3	13.0	189	7.3	74.0	−1.4	3/7	6.3	13.0	189	7.3	74.0	−1.4	36
KA 26	*7/9*	*−0.2*	*45.5*	*158*	*4.5*	*84.4*	*−160.5*	*7/9*	*−0.2*	*45.5*	*158*	*4.5*	*84.4*	*−160.5*	*37*
KA 25	6/6	0.1	30.0	88	6.5	84.7	21.1	6/6	0.1	30.0	88	6.5	84.7	21.1	38
KA 24	*5/7*	*−0.8*	*23.4*	*18*	*16.7*	*80.8*	*26.4*	*5/7*	*−0.8*	*23.4*	*18*	*16.7*	*80.8*	*26.4*	*39*
KA 23	4/5	5.9	17.7	183	5.9	76.4	−4.0								
KA 22	8/9	6.2	18.3	175	4.3	76.6	−5.7	12/14	6.6	17.9	237	3.95	76.3	−6.8	40
KA 21	*7/8*	*15.3*	*23.3*	*86*	*6.8*	*72.7*	*−38.4*								
KA 20	*5/6*	*12.2*	*18.7*	*88*	*7.3*	*73.3*	*−25.0*								
KA 19	*7/10*	*12.6*	*19.1*	*858*	*1.9*	*73.3*	*−26.4*								
KA 18	*7/7*	*8.4*	*20.0*	*95*	*5.7*	*76.3*	*15.4*	*26/31*	*12.0*	*19.7*	*131*	*2.6*	*73.9*	*−25.9*	*41*
KA 17	9/9	−10.6	33.2	97	5.0	79.5	95.9	9/9	−10.6	33.2	96.9	5.0	79.5	95.9	42
KA 16	*7/7*	*1.3*	*29.9*	*92*	*5.8*	*84.3*	*8.8*								
KA 15	*7/7*	*−2.4*	*29.2*	*203*	*4.2*	*83.7*	*43.7*	*14/14*	*−0.6*	*29.4*	*143*	*3.3*	*84.3*	*27.3*	*43*
KA 14	5/7	−4.2	19.0	136	6.8	77.7	41.7								
KA 13	7/10	−4.6	26.8	64	7.0	81.6	53.6	12/17	−4.5	23.9	77	5.2	80.1	48.6	44
KA 12	*7/8*	*−0.3*	*14.5*	*200*	*4.4*	*76.0*	*23.0*								
KA 11	*7/7*	*−0.2*	*14.1*	*115*	*5.2*	*75.7*	*22.6*								
KA 10a	*13/13*	*−3.5*	*14.2*	*187*	*2.9*	*75.4*	*35.7*	*27/28*	*−1.9*	*14.4*	*176*	*2.2*	*75.8*	*29.4*	*45*
KA 10	8/12	10.4	12.2	639	2.1	71.7	−13.1	8/12	10.4	12.2	639	2.1	71.7	−13.1	46
KA 09	*6/7*	*8.9*	*5.7*	*332*	*3.7*	*69.4*	*−4.4*								
KA 08	*4/6*	*12.8*	*7.4*	*137*	*6.8*	*68.3*	*−14.9*								
KA 07	*5/7*	*10.8*	*11.3*	*126*	*6.1*	*71.2*	*−13.5*								
KA 06	*4/8*	*7.9*	*13.7*	*81*	*8.9*	*73.6*	*−7.1*	*19/28*	*10.2*	*9.6*	*140*	*3.0*	*70.8*	*−10.5*	*47*
KA 06bis	4/8	11.5	−13.6	160	6.3	59.5	−1.1	4/8	11.5	−13.6	160	6.3	59.5	−1.1	48
KA 05	*5/7*	*34.7*	*−25.9*	*398*	*3.4*	*41.1*	*−25.5*	*5/7*	*34.7*	*−25.9*	*398*	*3.4*	*41.1*	*−25.5*	*49*
KA 04	7/8	−3.3	1.9	250	3.5	69.3	31.3	7/8	−3.3	1.9	250	3.5	69.3	31.3	50
KA 03	*5/6*	*−17.9*	*5.9*	*231*	*4.5*	*64.6*	*67.7*	*5/6*	*−17.9*	*5.9*	*231*	*4.5*	*64.6*	*67.7*	*51*
KA 02	9/9	3.8	37.5	1165	1.4	86.4	−62.2	9/9	3.8	37.5	1165	1.4	86.4	−62.2	52
KA 01	*9/9*	*−15.7*	*4.1*	*144*	*4.1*	*65.3*	*−62.3*	*9/9*	*−15.7*	*4.1*	*144*	*4.1*	*65.3*	*62.3*	*53*
KA 51	6/8	−6.5	19.3	97	6.2	77	51.3	6/8	−6.5	19.3	97	6.2	77.0	51.3	54

Alternating italicized and normal type indicate the clusters identified by the modified Vandamme and Bruneton method [29]. For each cluster the characteristic direction was obtained by averaging all the samples using Fisher's statistics.

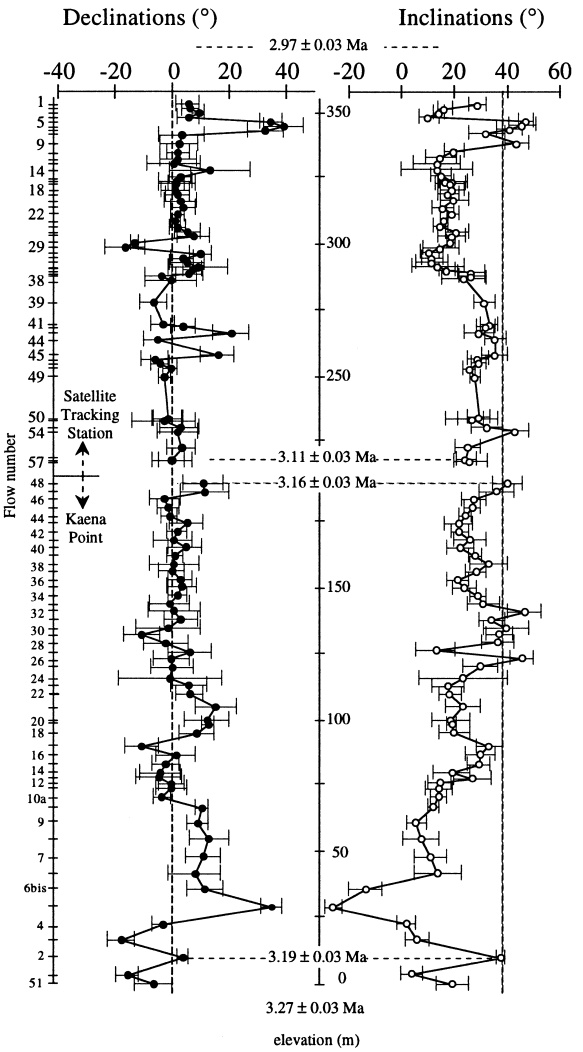


Fig. 4. Inclination and declination records as a function of flow numbers in stratigraphic order. Radiometric dates and their corresponding altitudes are also reported. Dashed lines show the declination and inclination expected from a geocentered axial dipole field at this latitude.

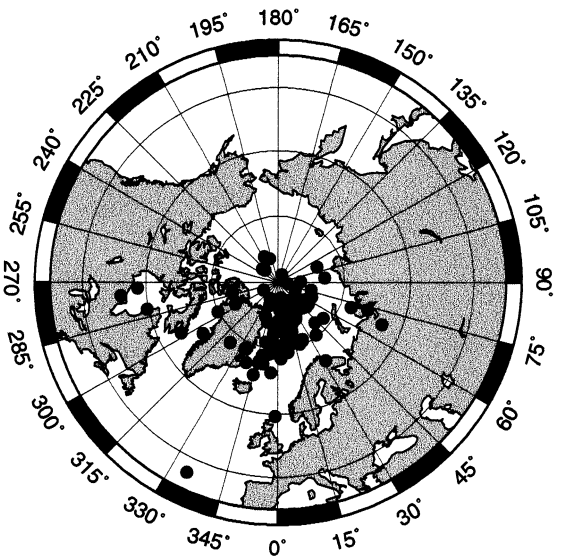


Fig. 5. Polar plots of the VGPs showing a clear far-sided effect.

Ma [26]) should have been at least partially recorded in the upper part of the section, while the paleomagnetic data do not document this event. This indicates that the real errors in dating may be slightly greater than those estimated by analytical precision.

Errors may arise from incomplete extraction during analysis, from inherited argon, or from element migration due to weathering. Because of the reproducibility of the data from independent aliquots, we are fairly certain that all the argon has been extracted during analysis. On the other hand, excess Ar and element migration are very likely candidates for explaining the inconsistencies between dates and paleomagnetic data. Excess $^{40}Ar^{*}$ resulting from incomplete degassing of magmatic Ar has been documented in some Hawaiian lavas [27]. Loss of radiogenic argon (Ar^{*}) which can result from devitrification of glass or other alteration processes has also been documented [27]. Minor K-losses have also been evidenced in the Waianae range from chemical analyses [15]. All these factors may lead to radiometric determinations either slightly younger or slightly older than the real age.

Despite these difficulties, a reasonably precise temporal accuracy can be obtained by combining the radiometric dates with the paleomagnetic results. As mentioned above, the Kamaileunu–Palehua contact is not encountered in the section. For the Waianae range, this boundary coincides with the Upper Kaena transition [15], so that the upper flows of the section are certainly older than this transition. Furthermore, because all the lavas yield normal polarities, the upper limit of our sampling is close but lower in stratigraphy than the lower Kaena transition (3.11 Ma [26]).

In the lower part of the section, two flows (KA05 and KA06bis) have reversed inclinations with de-

clinations only slightly deviated from the north and the corresponding VGPs are in the Northern Hemisphere. The K/Ar age of flow KA02 (3.19 ± 0.03 Ma) is consistent with its observed normal polarity. A flow situated at 0 m altitude and dated at 3.27 ± 0.03 Ma (sample OAU 06 [16]) was unfortunately not sampled for the paleomagnetic study and its polarity is unknown. Based on the radiometric dating of flow KA02 and on the absence of an interval of clear reversed paleomagnetic directions, we suggest that the KP–STSR sequence was emplaced entirely in the normal polarity chron between the Upper Mammoth transition (3.22 Ma) and the Lower Kaena transition (3.11 Ma). The two anomalous directions recorded by flows KA05 and KA06bis may represent either an episode of large secular variation, or possibly, as the low magnetizations of the neighboring flows may suggest, a short, so far undetected geomagnetic event.

Assuming that the section encompasses the entire interval from 3.22 to 3.11 Ma, one gets an estimate of about 4 ± 1.5 mm/year for the lava accumulation rate, which is a very realistic figure for the Waianae volcano [16]. In terms of flows per unit time (which is the relevant figure in paleomagnetic records) one gets a value of about 1 flow/1000 years averaged on the entire section. The radiometric data further suggest that this rate was much higher during the emplacement of the STRS than during the emplacement of the KP sequence. Unfortunately, attempts to obtain radiometric dates at intermediate horizons of the STRS failed for several reasons discussed elsewhere [16]. This lack of intermediate dates does not allow us to obtain a quantitative evaluation of the changes in extrusion rate through the entire sequence.

4.2. Serial correlation between flows

Because of the sporadic nature of volcanic extrusion, it may happen that a large number of flows are emitted in a time which is short compared to the rate of change of the field. Although numerous, these flows may then only provide a few independent readings of the geomagnetic field. If ignored, this serial correlation may lead to an incorrect estimate of the dispersion of the field direction.

Different approaches to this issue have been reported in the literature. In their study of the Steens Mountain sequence, Mankinen et al. [28] used direct comparison of successive flows. If the mean magnetization direction of two or more flows showed no systematic trends and their ovals of 95% confidence overlapped, they were considered to record the same field direction and averaged into a single directional group. If there was an overlap of confidence ovals but the mean direction of one flow progressed distinctly toward that of the following directional group, it was considered as a distinct direction. In general there were very few ambiguous cases.

More recently, McElhinny et al. [12] and Vandamme and Bruneton [29] have proposed specific statistical approaches to evaluate the extent to which the different flows yield independent observations. In both methods a critical angle is defined, so that when the angular difference of the paleomagnetic vector between two flows exceeds this critical angle, then it is considered that the two flows record different values of the geomagnetic field and vice versa. The two methods differ in the way the critical angle is calculated. In the Vandamme and Bruneton calculation, the critical angle depends specifically on the accuracy of the directional determinations of the two flows whose correlation is being tested. In the McElhinny et al. method, the critical angle results from an average over the entire section. A very different view has been held by Love [30], who remarked that limited changes in field direction between successive flows often correspond to changes in field intensity of up to tens of microtesla and thus represent independent readings of the field. Such intensity changes may arise from secular variation and occur on a time scale of centuries or less.

Given these different views, we have considered both unfiltered and filtered data to evaluate the impact that filtering may have on the geomagnetic conclusions drawn from the data. To assess serial correlation, we have used a slight modification of the Vandamme and Bruneton method: in each cluster composed of three or more flows, the test was repeated for each pair of flows within the cluster. This procedure avoided grouping together flows having recorded slow geomagnetic directional trends.

Using this modified method, the 103 flows from which reliable results were obtained are grouped into 54 directional clusters, shown as alternating itali-

cized and normal type rows in Table 2. The mean paleomagnetic direction of each cluster, shown on the right hand side of this table, was calculated with Fisher's statistics using all the samples in the cluster. The very large majority of the clusters are formed by two flows, showing that filtering does not drastically affect the characteristics of the record. In the upper part of the section, on the other hand, filtering identifies a very large cluster of twelve flows. If the methods in [12] and [29] had been used, the results would have been nearly the same; the three methods agree on limited serial correlation, with the exception of the large cluster in the upper part of the section identified in all three cases. This large cluster indicates that the extrusion rate was particularly high during emplacement of this part of the section, consistent with the radiometric dates. Because of these two independent criteria, we believe that this cluster is real, i.e. that it reflects high extrusion rate and consequent multiple sampling of the geomagnetic field.

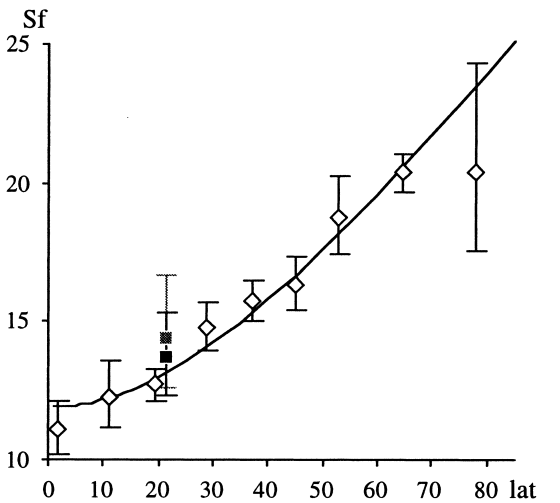


Fig. 6. The scatter of the VGP (with respect to the spin axis and corrected for the within-flow angular dispersion) is reported on the recent world compilation plotted as a function of latitude. Black (gray) square symbols: unfiltered (filtered) data. In both cases the large upper cluster is grouped (see text). Open diamonds are relative to the recent McElhinny and McFadden database [33] and the solid line represents the predictions of Model G [34].

4.3. Analysis of paleosecular variation

In the analysis of paleosecular variation we have discarded flow KA05 whose mean direction differs by more than 35° from the axial dipole [25], and flow KA24 characterized by within-flow scatter of 16.7°, as it is usually done in this kind of analysis. We have also assumed that the upper cluster is real, as discussed above, but for the rest of the section unfiltered and filtered data were both considered, in order to asses the impact of filtering on the interpretation of the results. This results in 52 independent readings of the geomagnetic field when filtering is considered.

Following McElhinney et al. [12], we have evaluated the scatter of the VGPs about the spin axis. The results are shown in Fig. 6 for unfiltered and filtered data. Within error limits the two values are indistinguishable and they are consistent with the global paleosecular variation database and with Model G of McFadden et al. [32,33].

We have also evaluated the secular variation by calculating the dispersion of the directions about their mean value. We have then compared this dispersion with the predictions of the Constable and Parker model [31], by generating time sequences using the time-dependent Constable model [34], in which we used the Gauss coefficient values from [25] and correlation times from [35]. We note that Kono and Tanaka [36] have shown that the Constable and Parker model does not give the observed worldwide increase in VGP dispersion with latitude and demonstrate how the variance of certain Gauss coefficients could be changed to make it do so. Nonetheless, the simplicity of the Constable and Parker model is appealing and is quite suitable for our purposes, because within 20° of the Equator the two models make very similar predictions.

We first generated sequences of 93 and 52 independent directions (consistent with the Waianae unfiltered, except for the large upper cluster, and filtered records). The directional scatter with respect to the mean direction was then calculated for five independent drawings, and the average value was considered to be a realistic estimation of a synthetic directional scatter. Fig. 7 shows that, for both unfiltered and filtered data, the directional dispersion obtained from the sequences is quite consistent with the predictions of the model, as illustrated by the very similar size of the confidence ellipses for the

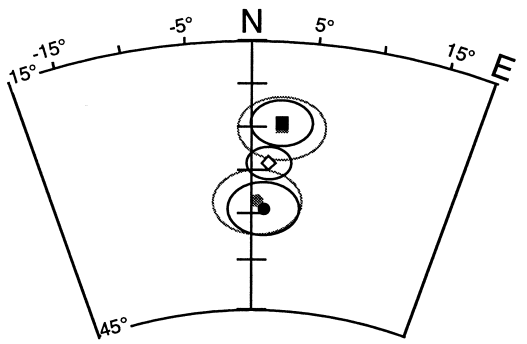


Fig. 7. Mean direction and confidence ellipse for the KP–STSR sequence, synthetic sequences (compiled from [31], see text), and previous normal Hawaiian data [25] (open diamonds). Black and gray square symbols as in Fig. 6, black and gray circles are for the model.

synthetic and Waianae mean directions. This visual consistency is supported by an F-test performed on the dispersion parameter K of the two directional distributions.

It is also apparent, however, that there is a significant difference in the directional mean which is entirely due to the inclination. The mean inclination retrieved from the KP–STSR, $25.3^{o} \pm 3.8^{o}$, is significantly shallower than that expected for a centered dipole field (38.2º). Although not inconsistent with all values observed from other sites at the same latitude (see e.g. [25,37]), as shown in Fig. 8, the inclination anomaly is among the largest observed for a large data set, and it is about twice the average for all flows from the Hawaiian Islands. This suggests that significant differences from the time-averaged field are sometimes maintained for periods in excess of 100 ka. Alternatively, one might tentatively suggest that part of the inclination anomaly has resulted from a slight (≈5–7º) southward, post-shield tilt of the Waianae volcano. Unfortunately, there is no supporting or contradictory evidence for this hypothesis (J. Sinton, pers. commun., 1998). Whatever the reason, however, this difference in inclination does not change the main conclusion of this study, namely that the amplitude of the secular variation around 3 Ma at Hawaii is quite consistent with values from other parts of the world.

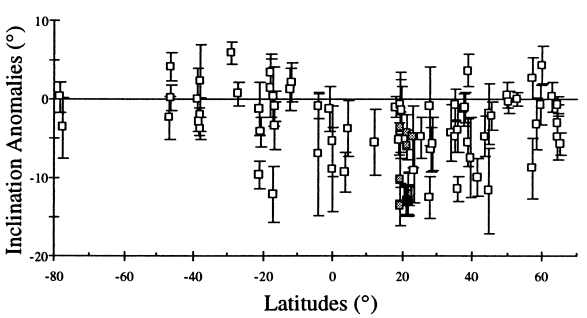


Fig. 8. Inclination anomalies as a function of latitude from [36]. Black and gray squares are respectively associated to this study and previous Hawaiian data compiled in [25].

5. Conclusions

This study presents a paleomagnetic investigation of a section of the Waianae volcano comprising two sections of successive lava flows, totaling 107 flows. Radiometric dates and paleomagnetic results provide an unusually well-documented record of secular variation over about 100 ka, in the normal polarity interval between the Upper Mammoth and the Lower Kaena polarity transitions. This length of time proved to be long enough to sample the secular variation adequately. The consistency of the VGP scatter and of the directional scatter with available data from other geographical areas and with modern models of the secular variation does not support the hypothesis of a Pacific dipole window. This conclusion, already reached from reexamination of old data [12], is obtained here from an entirely new study combining radiometric dating with modern paleomagnetic laboratory methods.

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